

Micro I, class 17

For $\succsim \subset \mathcal{L} \times \mathcal{L}$ define the certainty equivalent to be the number $CE(F)$ satisfying

$$F \sim \delta_{CE(F)}$$

and the risk premium to be $RP(F) = \int z dF(z) - CE(F)$. Consider two people with expected utility preferences represented by C^2 vN-M utilities with positive first derivatives. The Arrow-Pratt theorem asserts a person's risk premium for a given risk is always at least as large than another's if and only if the first person's Arrow-Pratt risk aversion measure is always at least as large as the second's. The formal statement is in my notes on the Arrow-Pratt theorem.¹

We then turned to an implication that the demand for a risky asset (in our two-asset portfolio problem with one safe and one risky asset): an increase in risk aversion *does not increase* the demand for the risky asset. The details are in my notes on Pratt's theorem on risk aversion and portfolio demand.

I mentioned that this rather-unsurprising-sounding conclusion is fragile. For example, if there are *three* assets, two risky and one safe, a natural conjecture would be that an increase in risk aversion *does not decrease* investment in the *safe* asset (while remaining silent about how wealth is allocated among the two risky assets). This natural conjecture is false.² Other seemingly-natural conjectures are false.³

I next will introduce the notion of an *increase in risk*. For F, G in $\mathcal{L}$, G is riskier than F if

$$\int f(z) dG(z) \leq \int f(z) dF(z) \quad \text{for every continuous, concave } f \text{ on } \mathbb{R}_+.$$

The definition implies that the variance of G is greater than or equal to the variance of F . The converse is false (even if we restrict attention to cdfs with the same mean).

The final issue for choice theory under uncertainty is a *subjective expected utility* (SEU) representation. I'll briefly cover the Aumann-Anscomb theorem on a SEU representation theorem. In this version of expected utility, preferences are not over probability distributions over outcomes, but over *acts*, functions from unknown states of the world, a finite set S , into consequences. A conclusion of the theorem is that, under some axioms, there is a vN-M utility u and a probability distribution π on S such that act y is at least as good as act x if and only if

$$\sum_{s \in S} u(y_s) \pi(s) \geq \sum_{s \in S} u(x_s) \pi(s).$$

I have notes on this theorem on Canvas (Subjective Expected Utility and Beliefs).

¹MWG's version of this theorem is Proposition 6.C.2. They include two additional equivalences that my version does not. Their condition (v) is especially interesting, since it doesn't rely on expected-utility preferences. A good treatment of it is in Yaari, Menahem E. "Some remarks on measures of risk aversion and on their uses." *Journal of Economic Theory* 1.3 (1969): 315-329.

²Hart, Oliver D. "Some Negative Results on the Existence of Comparative Statics (Properties) in Portfolio Theory." *The Review of Economic Studies* 42.4 (1975): 615-621.

³I know whereof I speak. My dissertation was on extending Arrow-Pratt comparative statics to situations in which there is more than one risk. That's not what I intended to write on, but, to do what I thought I needed to do to answer my bigger question, I felt—wrongly, as it turns out—that I needed to answer this question about multivariate risk aversion.